2025 International Conference on Electronics, Computing, Communication and Control Technology (ICECCC) | 979-8-3315-2162-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICECCC65144.2025.11064291

Predictive maintenance system for CNC machines

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***Abstract*—The present invention relates to BE SHIELD, a state-of-the-art preventive maintenance system specifically designed for CNC machines in the hardware manufacturing industry. This invention addresses a critical niche, improving the operational efficiency and reliability of industrial machinery through advanced predictive maintenance. It comprises an integrated sensor kit installed on a CNC machine, equipped with various sensors, including temperature, pressure, and vibration sensors. The real-time data collected by these sensors is transmitted to a maintenance software. This software employs a sophisticated artificial intelligence model to monitor and analyses parameter trends continuously. By accurately predicting potential breakdowns in advance, this invention allows for timely pre-emptive actions, significantly reducing the risk of unexpected machine failures, downtime, and maintenance costs. In an industry, where expert maintenance personnel are not always readily available, this invention offers a substantial advantage. It ensures that experts can be scheduled proactively, thus enhancing the overall efficiency and productivity of manufacturing operations.**

Keywords— artificial intelligence model, CNC machines, predictive maintenance, real-time data, sensors, , breakdowns, efficiency

I. Introduction

”Be Shield” is an advanced predictive maintenance system developed to enhance the efficiency and reliability of CNC (Computer Numerical Control) machines, which are integral to modern manufacturing. The system incorporates a sensor kit featuring an Arduino Uno to gather real-time data from vibration, temperature, and humidity sensors. By utilizing machine learning algorithms, this data is analysed to predict potential breakdowns and maintenance needs. These predictions are integrated into a maintenance management platform, which provides real-time alerts and actionable insights, enabling proactive interventions that minimize downtime and optimize machine performance and lifespan.

CNC machines are automated manufacturing tools that utilize computer-controlled operations to achieve high precision and efficiency. Common types of CNC machines include mills, lathes, plasma cutters, and grinders, which are employed across industries such as aerospace, automotive, healthcare, and electronics. Their ability to create complex, uniform designs with minimal defects has revolutionized manufacturing processes, significantly enhancing productivity and reducing errors. However, their high initial investment and reliance on skilled operators make effective maintenance crucial for sustaining performance and cost-efficiency. It is important that the machines work uninterruptedly as planned and that there is no downtime due to failures for the companies to survive in tough competitive conditions and to deliver the production with the commitment ethics on time. An unforeseen failure can result in a catastrophic accident or a failure in the system may cause a downtime resulting in financial losses[1].

Traditional maintenance approaches for CNC machines, including reactive and preventive strategies, face significant limitations. Field failures related to the operating environments of electronic equipment show that about 55% of the failures are due to high temperatures and temperature cycling, twenty percent are related to vibration and shock, and another twenty percent are due to humidity[2]. Reactive maintenance, performed after machine failure, often results in unplanned downtime, costly repairs, and cumulative component damage. Preventive maintenance, while scheduled proactively, is typically based on general usage patterns and fails to account for specific operating conditions, leading to over- maintenance or under-maintenance. These inefficiencies not only disrupt production but also contribute to increased operational costs and resource misallocation. The limitations of traditional maintenance approaches have significant operational implications. Reactive maintenance causes extended downtimes and escalated repair costs due to emergency interventions and potential replacement of critical components. Meanwhile, preventive maintenance can waste resources through unnecessary servicing or leave critical issues unresolved. These challenges underscore the need for data-driven, predictive maintenance systems like ”Be Shield,” which leverage sensor-based monitoring and machine learning

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analytics to deliver tailored, condition-based maintenance strategies. Early diagnosis and prediction of fault can prevent the critical system failures and downtime. Efficient maintenance approaches, particularly PdM of machinery, manufacturing systems, production processes have emerging importance in predicting overall life cycle of machinery and machining tools [3][4].

The primary objective of this research is to design a universally compatible sensor kit, that can be seamlessly integrated with any CNC machine improving shop floor management, and decrease downtime by predicting occurrence of critical equipment failures[5], regardless of its brand or model. The sensor kit is capable of collecting real-time data on critical machine parameters such as temperature, humidity, vibration, and emergency stop signals, providing a comprehensive overview of the machine's operational condition. Through data acquisition from machines and manufacturing systems, IoT platforms will support PdM [6] This system is aimed at enabling robust, adaptable solutions for a wide range of manufacturing environments, ensuring flexibility and broad applicability across industries. Another key objective is to facilitate seamless data transmission for remote monitoring and analysis. By leveraging machine learning predictions, the system provides timely and actionable maintenance alerts, allowing teams to address potential issues before they escalate into significant failures. Machine learning, a subcategory of AI, is a modeling and algorithm method that extrapolate from measured or available data by means of mathematical and statistical methods and accurately predict the unknown. Various ML algorithms to make predictions are developed by researchers which are categorized as unsupervised learning and supervised learning. Unsupervised learning methods are utilized to find unknown structures and relationships in the data and model the distribution to find similar sample groups in the data [7] Additionally, a web-based interface has been developed to centralize the monitoring process, enabling maintenance teams to view real-time sensor data, analyse machine learning predictions, and manage maintenance activities for multiple machines in one platform. This interface enhances decision-making efficiency by providing clear and actionable insights, ensuring a proactive approach to CNC machine maintenance. The Be Shield system has been designed primarily on CNC milling machines, which are commonly used in small- to medium-scale manufacturing. However, the sensor kit and machine learning pipeline are built to be modular and adaptable, making the system generalizable to other CNC types such as lathes, plasma cutters, and grinders. This flexibility is ensured by using universally applicable sensor parameters like vibration, temperature, and humidity that are critical across machine types.

II. PREDICTIVE MAINTENANCE SYSTEM

A. Hardware and software components

The "Be Shield" system is a comprehensive predictive maintenance solution for CNC machines, integrating advanced hardware and software components to ensure optimal machine performance and minimal downtime.The hardware framework begins with a vibration sensor module, a piezoelectric sensor capable of detecting frequencies from 0.5 Hz to 40 Hz, with a sampling rate of 100 Hz. This makes it particularly suitable for identifying short-duration anomalies, such as misalignments, wear, or structural imbalances. Its high sensitivity enables the detection of micro-level vibrations, offering early insights into potential mechanical failures. Additionally, the system employs the DHT11 humidity and temperature sensor module to monitor environmental conditions. The DHT11 records temperature (0°C to +50°C) and humidity (20% to 90% RH) at 5-second intervals, appropriate for capturing slower environmental changes. This environmental monitoring helps reduce risks like thermal stress, corrosion, and moisture-related damages that can compromise machine longevity. Central to the system is the Arduino UNO R3 microcontroller, a reliable and versatile device equipped with the ATmega328P processor. It serves as the primary processing unit, collecting real-time data from the connected sensors. Data is collected through the analog and digital pins on the Arduino UNO R3, aggregated every second for vibration readings and every 5 seconds for environmental readings, then transmitted via USB serial communication to the Python-based software layer for further analysis.

On the software front, the Arduino IDE provides a platform to write, compile, and upload code for sensor management and data transmission. Python, known for its simplicity and extensive libraries, forms the backbone of data analysis and predictive modeling. The train_data.py and test_data.py scripts generate synthetic datasets mimicking real-world sensor outputs, incorporating parameters like temperature, vibration, pressure, and machine breakdown labels. These datasets are stored as CSV files using the Pandas library, providing the foundation for training and testing the predictive maintenance algorithms. Scikit-learn's Random Forest Classifier is used to develop a robust model capable of predicting breakdowns based on sensor input features. Random Forest trains multiple decision tree models using both the sample dimension and the feature dimension. By integrating the voting results of multiple decision trees, it addresses the common issue of decision tree overfitting [8]. In our study, this ensemble model was implemented using the Scikit-learn library from Python 3.9.16 Release [9]. This predictive model evaluates incoming data during real-time simulations, with Python's time module simulating live monitoring by introducing delays between predictions. The system visualizes insights through Matplotlib, creating intuitive graphs and charts that showcase feature importance and the model's predictive performance.

To enhance user accessibility, a centralized web-based dashboard consolidates all data into an intuitive interface. This platform displays real-time sensor readings, predictive analytics, and actionable maintenance alerts. By presenting

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operational parameters in visual formats such as line graphs and comparison charts, the interface empowers maintenance teams to monitor multiple machines simultaneously and prioritize interventions effectively. The integration of advanced hardware and intelligent software ensures seamless data acquisition, robust analysis, and efficient communication, making ”Be Shield” an innovative and essential tool for modern CNC machine maintenance.

B. Methodology

The proposed system adopts a comprehensive methodology for predictive maintenance of CNC machines, beginning with real-time data collection from an integrated sensor kit. This kit includes the DHT11 sensor for monitoring temperature and humidity, and a vibration sensor module for detecting mechanical anomalies. These sensors continuously gather high-resolution data on the machine’s operational parameters, offering real-time insights into its condition. The Arduino Uno R3 acts as the central processing unit, aggregating data from the sensors and applying local pre-processing techniques such as noise filtering and data structuring. This ensures that clean, actionable data is transmitted for analysis, reducing latency and improving system efficiency. The raw sensor data undergoes several preprocessing steps before being fed into the machine learning model. Data cleaning includes removing outliers and filling missing values using interpolation or rolling averages. Normalization is performed using Min-Max scaling to standardize feature ranges between [0,1], ensuring that no single feature dominates due to scale. Feature extraction includes calculating rolling statistics (mean, standard deviation, and root mean square (RMS)) for vibration signals, which are more informative than single-point measurements. These extracted features provide a temporal dimension to fault detection.

The collected data is analyzed using a machine learning model developed with Scikit-learn’s Random Forest Classifier. The Random Forest Classifier was chosen due to its ability to handle high-dimensional data, reduce overfitting through ensemble voting, and provide interpretability via feature importance scores. While classification models predict the likelihood of failure, future extensions may explore regression-based models such as Gradient Boosting Regressors for estimating Remaining Useful Life (RUL). This model, trained on historical data, identifies patterns and relationships between sensor inputs and machine breakdowns. Instead of creating physical or statistical models in the industrial domain, AI approaches aim to learn degradation patterns of machines from the data currently available. They can handle complex mechanical system prognostic issues whose degradation processes are challenging for physical or statistical models to relate to one another [10]. The Random Forest algorithm, which combines the output of multiple decision trees to enhance predictive accuracy, plays a crucial role in detecting anomalies. The Decision Tree algorithm generates a decision model that classifies instances by sorting the feature space into subspaces starting at the root node until a termination criterion is met [11][12]. Real-time sensor readings are compared with historical patterns to identify irregularities, such as abnormal vibration levels, temperature spikes, or humidity fluctuations, that may indicate impending failures. Feature importance analysis using the Random Forest model shows that temperature, vibration level, and pressure are the most influential parameters in predicting machine failures. These features exhibit strong correlations with key failure modes: high vibration levels often indicate issues such as misalignment, tool wear, or spindle bearing faults; a rise in temperature typically points to overheating problems in motors or control systems; and fluctuations in humidity levels are associated with corrosion risks or sensor malfunctions.

By identifying these relationships, the feature selection process ensures that the model’s predictions are grounded in real-world failure mechanisms, enhancing both the accuracy and practical relevance of the predictive maintenance system. Additionally, the system incorporates a feedback loop, continuously retraining the model with new data to adapt to evolving machine behaviour and improve predictive accuracy over time. When anomalies are detected, the system generates pre-emptive breakdown alerts. These alerts, which include details about the nature of the anomaly and the predicted time of failure, enable maintenance teams to address issues proactively, preventing significant disruptions. The alerts are categorized based on severity, allowing teams to prioritize critical tasks and allocate resources efficiently. Alongside alerts, the system provides actionable maintenance recommendations, such as recalibrating sensors, checking for structural misalignments, or replacing components at risk of failure. This feature ensures that maintenance actions are targeted and effective. A user-friendly real-time interface enhances the system’s accessibility, displaying sensor readings, breakdown predictions, and maintenance alerts through intuitive visualizations. Graphs and charts generated using Matplotlib, such as time-series plots and feature importance rankings, provide clear and actionable insights into machine health. This centralized interface empowers maintenance teams to monitor multiple machines simultaneously, facilitating informed decision-making and improving overall operational efficiency.

The methodology emphasizes a proactive maintenance approach by tracking and managing tasks in real time. Alerts and their resolutions are logged, ensuring a systematic record of all maintenance activities. This organized workflow not only minimizes downtime but also extends the lifespan of CNC machines. Through continuous data collection, real-time feedback integration, and actionable insights, the system ensures robust and adaptive predictive maintenance, helping industries maintain uninterrupted production cycles while optimizing resource utilization.

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III. PROPOSED SYSTEM ARCHITECTURE

Fig.1 depicts the exact structure of the Be Shield Sensor Kit (104). The sensor kit is attached to a certain CNC machine (101), and it includes several critical components. At its heart is the Arduino UNO R3 (110), which serves as the main processing unit. The Memory Element (111) manages data storage, while several sensors monitor the CNC machine's operational

environment, including the Temperature Sensor (112), Humidity Sensor (113), and Vibration Sensor (114). These sensors monitor the conditions inside the CNC machine, providing critical data for study. The collected data is displayed on Display module(115). The Power Supply (116) powers the whole sensor kit package, ensuring that it operates consistently and reliably. The data is subsequently processed by the Be Shield Maintenance Software, which provides superior predictive maintenance capabilities.

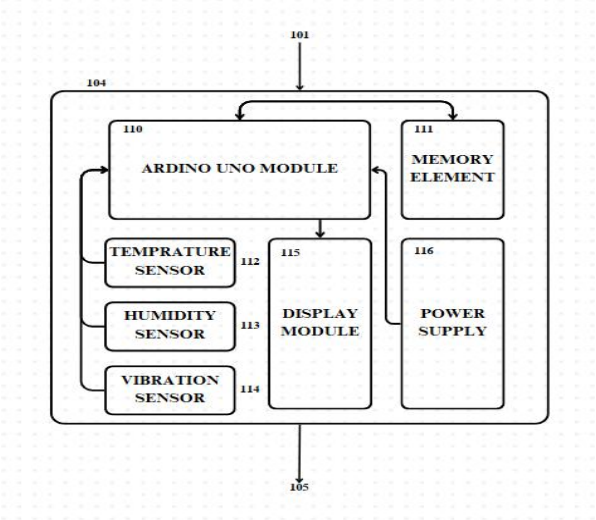

Fig. 1. Detailed Structure of Be Shield Sensor Kit

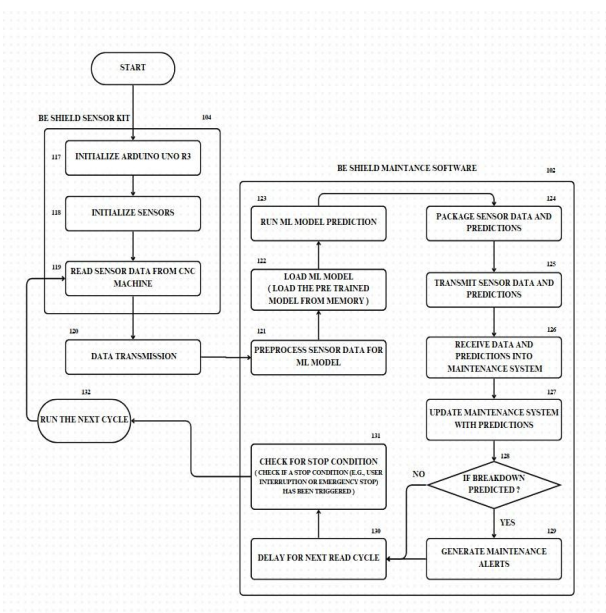

Fig. 2. Operational Process flow chart of Be Shield system

Fig.2 provides a full picture of the Be Shield system's operating workflow, which includes sensor data gathering, machine learning (ML) predictions, and CNC maintenance updates. The approach starts with initializing the Be Shield sensor kit (104), then the Arduino UNO R3 (117) and

sensors (118). The system next reads sensor data from the CNC machine (119) in real time before transmitting the data (120). The sensor data is pre-processed for the machine learning model (121), which is then loaded from memory (122). The ML model predictions are executed (123), and the sensor data and predictions are packed (124) and uploaded (125). The maintenance system accepts data and predictions (126) and updates itself as needed (127). If a breakdown is predicted (128), maintenance alerts will be given (129). If no breakdown is predicted, the system waits until the next read cycle (130). Throughout the procedure, the system monitors for stop situations such as user interruptions or emergency stops (131). Repeating the cycle (132) ensures continuous tracking and predictive maintenance of the CNC machine. The pin diagram is shown in Fig.3.

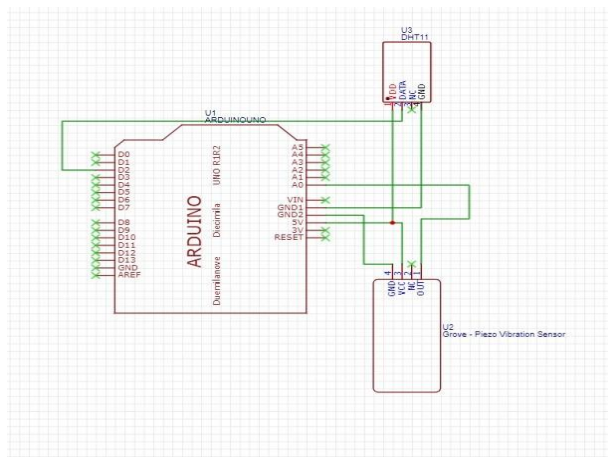

Fig. 3. Pin Diagram

IV. EXPERIMENTAL RESULTS

To evaluate the effectiveness and optimize the predictive maintenance system, we developed and tested four versions of the model. Each version incorporated iterative improvements in sensor integration, data processing, and machine learning algorithms to enhance prediction accuracy and system performance.

A. Version 1

This version implements a predictive maintenance solution for CNC machines using machine learning. Two Random Forest Classifiers are developed to predict: (1) the likelihood of machine breakdowns and (2) the need for servicing. The dataset includes features such as temperature, pressure, vibration levels, and machine age. The data is split into independent features and target variables (Breakdown and Service Needed), followed by training and testing subsets. The models are trained separately for each prediction task, achieving satisfactory accuracy. Supervised

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learning is used in most practical ML applications. An algorithm known as supervised learning begins with a training dataset that contains both the input and output variables[13] Predictions are also generated for a new dataset (test_set.csv), and the results, including

predicted breakdowns and service requirements, are saved for further analysis. In most MLSs, trained learning is used. The available data sets are referred to as “true” or “correct” data sets during the learned learning phase. These input data sets, also known as “training data,” are used to “feed” the algorithm[14]

The project incorporates feature importance analysis to identify key variables influencing the predictions. Bar charts highlight the most critical features for both prediction tasks, such as temperature, vibration levels, and wear and tear. Scatter plots as shown in Fig.4 visualize the predicted outcomes, showcasing how the models differentiate between breakdowns and service needs based on factors like temperature, pressure, vibration levels, and machine age. These insights enable effective prioritization of maintenance activities, ensuring timely intervention to prevent machine failures and optimize operational efficiency.

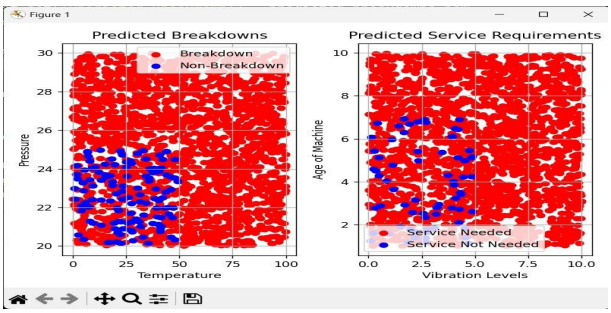

Fig. 4. Scatter Plot

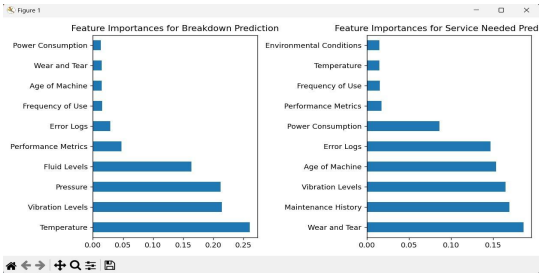

Fig. 5. Bar Plot

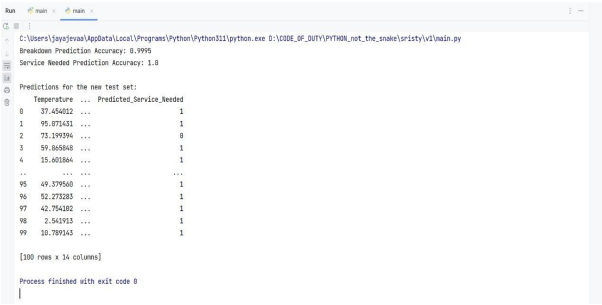

Fig. 6. Test Predictions

The scatter plots depict the distribution of prediction outcomes. In the first plot, red dots represent predicted breakdowns, clustering at higher temperature and pressure values, while blue dots (no breakdowns) are concentrated at lower ranges. In the second plot, red dots signify machines requiring servicing, primarily appearing at higher vibration levels and older machine ages, whereas blue dots (no service required) cluster at lower values for both parameters.

The bar charts as in Fig.5 highlight key features influencing predictions. For breakdown prediction, temperature, vibration levels, and pressure are the most critical factors. For service-needed prediction, wear and tear, maintenance history, and vibration levels are the top contributors. Other features have moderate to minimal impact on both models.

The output as shown in Fig.6 showcases the models’ exceptional reliability, with a Breakdown Prediction Accuracy of 99.95% and Service Needed Prediction Accuracy of 100%. Predictions for a new dataset highlight that machines with higher temperatures and vibration levels are more likely to require servicing. Results are saved in a CSV file, enabling actionable insights for maintenance scheduling and machine management.

B. *Version 2*

The output from the second version of the machine breakdown detection system highlights its enhanced real-time predictive capability. Utilizing a Random Forest Classifier, the system processes input features to categorize machine states as normal or breakdown predicted. Predictions are displayed sequentially, simulating real-time analysis and enabling immediate attention to critical alerts. as shown in Fig.7. A prompt feature ensures controlled progression, mirroring real-world operator interactions. This version effectively monitors equipment health, identifies potential breakdowns with high precision, and supports proactive decision-making, offering a practical solution for predictive maintenance and improved operational reliability.

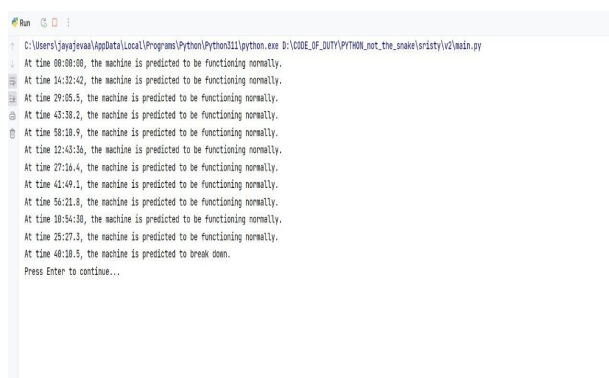

Fig. 7. Predictions based on real-time

C. *Version 3*

Version 3 introduces a Flask-based web application tailored for real-time monitoring and prediction of machine breakdowns. It integrates a Random Forest-based

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Fig. 8. Register page

Fig. 9. Dashboard

predictive maintenance model, leveraging historical data to evaluate machine health. The application features a secure and user-friendly interface for registering devices, monitoring their status, and receiving alerts for potential breakdowns. Essential functionalities such as user authentication, device registration, and a centralized dashboard enhance its utility, ensuring secure and efficient machine management as shown in Fig.8 -13.

At the core of the system is the simulate_real_time_prediction function, which uses the trained Random Forest model to perform real-time predictions for registered devices. This function iteratively processes incoming data, classifying machines as ”Normal,” ”Potential Breakdown,” or ”Breakdown.” Predictions are displayed on a dynamically updated dashboard, enabling users to track the operational health of multiple machines simultaneously. Real-time simulation is handled efficiently through threading, ensuring seamless functionality across devices without interrupting other application processes.

JavaScript enhances the system’s real-time monitoring by fetching machine statuses from the backend at one-second intervals. It dynamically updates the user interface

Fig. 10. Login Page

using asynchronous fetch requests, displaying details such as timestamps and operational statuses. Visual indicators like status dots—colored green, orange, or red—represent normal operation, potential issues, or critical breakdowns, respectively. The script also includes alert triggers for critical statuses, halting further updates when necessary. This visually intuitive and scalable design ensures effective monitoring, reducing downtime and supporting timely interventions for predictive maintenance.

D. Version 4

This is the fourth version of the system, which implements a Python script for real-time sensor data collection from an Arduino. The script establishes a serial connection with the Arduino (via COM6 at 9600 baud rate) to receive sensor data, including temperature, humidity, and vibration readings. The received data is parsed and adjusted (humidity increased by 30 and vibration by 200) before being passed to a pre-trained machine learning model. The model predicts whether a breakdown is likely, outputting a prediction (0 for no breakdown, 1 for breakdown predicted) along with the sensor values to the console. The script also logs the sensor readings, timestamp, and breakdown prediction to a CSV file (’real_time_data.csv’). If the file does not exist, it is created with a header, ensuring that the header is written only once to avoid duplication. The data is appended every 5 seconds, and the loop continues indefinitely, reading, processing, and logging the data until an error occurs as shown in Fig.14. If there is an issue with the serial connection or data parsing, the script catches the error and prints a message to the console, ensuring smooth real-time monitoring and logging without interruption.

V. CONCLUSION

In conclusion, the four versions of the predictive maintenance system showcase a progressive enhancement in monitoring and predicting machine breakdowns for CNC machines. The first version focused on real-time data collection using integrated sensor kits, where sensor readings like temperature, humidity, and vibration were continuously monitored. The second version introduced a Flask-based web application, allowing real-time monitoring, breakdown predictions, and service alerts through a user-friendly interface. The third version built on this by integrating real-time machine status updates, leveraging JavaScript for dynamic, asynchronous updates in the user interface. The fourth version incorporated a Python script for direct communication with an Arduino, collecting sensor data and utilizing a machine learning model for breakdown prediction, which was logged in real- time to a CSV file. Together, these versions demonstrate a scalable, reliable, and practical solution for predictive maintenance, combining sensor data, machine learning, real-time monitoring, and user-friendly interfaces to reduce downtime, improve operational efficiency, and facilitate proactive maintenance scheduling. Through continuous integration of new technologies and enhancements, the system ensures timely detection of potential issues,

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Fig. 11. Green Status operating normally

Fig. 12. Orange Warning - Breakdown likely

Fig. 13. Red Breakdown predicted

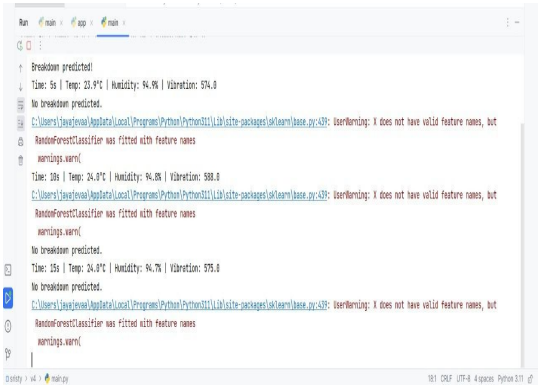

Fig. 14. Breakdown Prediction

ultimately contributing to enhanced machine reliability and prolonged operational lifespan.

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